

PRAVEEN PAL RAWAL

AI / ML Engineer

Jodhpur, Rajasthan, India | +91 85296 23327 | rawalpraveen886@gmail.com

 linkedin.com/in/praveen-rawal |  github.com/thepietro23

Education

M.Tech, Data Science & Machine Learning

2024 – 2026 (Expected)

Rashtriya Raksha University, Gandhinagar, Gujarat | CGPA: 8.52

B.Tech, Computer Science & Engineering (AI)

2019 – 2023

Vivekananda Global University, Jaipur, Rajasthan | CGPA: 9.1 | Silver Medalist (2nd Rank)

Experience

AI/ML Intern | *Forensic Cybertech*

Jan 2026 – Present

Built an EDR behavioral malware-detection model specializing in real-time **ransomware detection**, profiling malicious process behavior instead of relying on static signatures.

Engineered a **UEBA** (User & Entity Behavior Analytics) baseline using 5 ML models to learn normal user behavior; flagged and analyzed anomalous users deviating from baseline and triggered automated alerts.

Building **agentic AI** systems (autonomous LLM agents) to proactively safeguard the product against threats.

Containerized and deployed models on **AWS** using Docker; worked with Apache Kafka for streaming and LangChain for LLM workflows.

GenAI Backend Developer Intern | *Techity Solutions Pvt. Ltd.*

Oct 2025 – Jan 2026

Developed AI-powered backend systems and REST APIs integrating large language models (LLMs) for production-grade GenAI applications.

Designed and integrated LLM pipelines into backend services, translating complex technical requirements into reliable, deployable solutions.

Implemented and tested API endpoints for AI features with robust error handling and reliable LLM response processing.

AI Engineering Intern | *AI Core System*

Jun 2025 – Aug 2025

Engineered an end-to-end speech-to-email automation pipeline using Whisper ASR, LangChain, and FastAPI; cut manual reporting time by ~83%.

Added silence detection, input validation, and retry logic for noisy audio, achieving <2s end-to-end latency.

Integrated SendGrid API for templated automated dispatch of 50+ daily reports with error handling.

AI Workshop Trainer | *NCII SECEX, Rashtriya Raksha University*

Apr 2025

Delivered applied AI/ML workshops on cybersecurity use cases to 150+ professionals and government officials.

Designed curriculum on neural networks, anomaly detection, NLP for threat intel, and deployment with FastAPI & Docker.

Key Projects

ResearchSynth AI — Multi-Agent Research Platform | *LangGraph, RAG, FastAPI, React*

2025

Architected a 6-agent + critic system in LangGraph with conditional feedback loops for autonomous research synthesis.

Built a RAG pipeline (ChromaDB + sentence-transformers + Groq API) for <1.5s inference; deployed full-stack FastAPI + React with SSE streaming and multi-format export.

AI-Powered Log Processing Pipeline | *Kafka, ML, NER, Python, Elasticsearch*

2025

Designed a SIEM-grade self-healing log pipeline ingesting 30K+ logs/sec; 90% decode at 0.18ms/log via regex, rest via ML (NER + embeddings).

Built an Ollama (phi4-mini) + sentence-transformer auto-decoder (93.8% NER, F1 0.85); added 231-field schema, Isolation Forest anomaly detection, and MITRE ATT&CK enrichment.

Climate-Finance Risk Assessment | *PyTorch, TensorFlow, GANs, LLMs*

2025

Modeled multi-modal data (climate, financial time-series, ESG text) with GANs and transformer LLMs for probabilistic risk scenarios.

Built reproducible ML pipelines with validation, imputation, and dataset versioning; generated 2050 climate-risk scenarios for ESG strategy.

Technical Skills

ML / DL	PyTorch, TensorFlow, Transformers (Hugging Face), scikit-learn, XGBoost, LightGBM
LLMs & Agents	LangChain, LangGraph, RAG Pipelines, Prompt Engineering, Groq API, OpenAI API, LLaMA-2, Fine-Tuning
NLP & Speech	NLP, Whisper ASR, BERT, FinBERT, spaCy, NLTK, Sentence Transformers, Text Classification, NER
Computer Vision	YOLOv8, BLIP, Grad-CAM, Image Segmentation, Object Detection, MS-COCO
Data & Pipelines	Apache Kafka, Apache Spark, PostgreSQL, MySQL, ChromaDB, Elasticsearch, Redis, Vector Search (HNSW)
MLOps & Deploy	Docker, FastAPI, Streamlit, Gradio, GitHub Actions, CI/CD, REST APIs, MLflow, Model Evaluation
Languages & Tools	Python (Adv.), SQL (Adv.), Bash, Linux, Git, Jupyter, Power BI

Achievements & Leadership

Runner-up, Smart India Hackathon (National), 2022 — among 10,000+ teams; **Silver Medalist** (2nd Rank), B.Tech CSE (AI).

Certified — **Neural Networks & Deep Learning** (DeepLearning.AI) and **Big Data Fundamentals** (IBM), Coursera 2025.

Founder, AI Club (RRU) — applied ML sessions for 50–60 students; **Junior Under Officer, NCC** — led 120 cadets, NCC 'C' Cert. Grade A.

Liaison & Operations Coordinator, AIPSC (Nov 2024) — coordinated end-to-end logistics for a national conference with 200+ delegates, including senior IPS/IAS officials.